Circular Parking

Problem Description

The school festival is about to begin, and the school has opened a circular parking lot for visitors.

There are n parking spaces, numbered from 1 to n. Since the parking lot is circular, the next space after space n is space 1. Now m cars enter the parking lot one by one. The i-th car prefers parking space x_i. If space x_i is occupied, the car continues searching clockwise until it finds the first empty parking space. The search wraps around from space n back to space 1 if necessary.

It is guaranteed that $m \leq n$, so every car can always find an empty parking space. Help determine the final parking space of each car.

Input Format

The first line contains two integers n and m, representing the number of parking spaces and the number of cars.

The second line contains m integers $x_1, x_2, \ldots, x_m$, where x_i is the preferred parking space of the i-th car.

Output Format

Print m integers separated by spaces. The i-th integer should be the final parking space of the i-th car.

Constraints

- $1 \leq m \leq n \leq 200000$
- $1 \leq x_i \leq n$

Example Test Case

Sample Input 1

```
5 4
3 3 5 4
```

Sample Output 1

```
3 4 5 1
```

Explaination

The first car parks in space 3. The second car finds space 3 occupied and parks in space 4. The third car parks in space 5. The fourth car finds spaces 4 and 5 occupied, wraps around, and parks in space 1. Therefore, the final parking spaces are 3, 4, 5, and 1.

Sample Input 2

```
7 6
6 6 6 2 7 1
```

Sample Output 2

```
6 7 1 2 3 4
```